📄 **ASSIGNMENT REPORT**

Text Analytics – Sentiment Analysis

Student Details

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- **Course:** Data Analytics and Visualization (CSC601)
- **Class:** TE (AI & DS)
- **Semester:** VI

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1. Introduction

This project focuses on performing sentiment analysis on tweets related to the CDM tournament in Call of Duty Mobile (CODM). With the rapid growth of mobile esports, analysing public opinion about tournaments and gameplay has become important.

The main objective of this project is to classify tweets into three categories: positive, neutral, and negative sentiments using machine learning techniques. This helps in understanding user reactions and overall sentiment towards the tournament.

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2. Data Collection & Preprocessing

A total of 100 tweets related to the CDM tournament and CODM gameplay were collected manually from Twitter and similar sources. Due to the limited availability of tweets specifically related to CDM, general CODM tournament tweets were also included.

Each tweet was manually labelled into one of the following categories:

- Positive
- Neutral
- Negative

Preprocessing steps performed:

- Conversion of text to lowercase
- Removal of unnecessary characters and symbols

- Ensuring consistent formatting of text

This preprocessing helped improve the quality of data for model training.

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3. Data Splitting

The dataset was divided into two parts:

- **Training Set:** 80 tweets (80%)
- **Testing Set:** 20 tweets (20%)

The training dataset was used to train the models, while the testing dataset was used to evaluate model performance on unseen data.

This 80–20 split ensures a good balance between training and evaluation.

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4. Model Training & Classification

Three machine learning models were used for sentiment classification:

1. **Naive Bayes**
2. **Logistic Regression**
3. **Support Vector Machine (SVM)**

Before applying these models, text data was converted into numerical format using **CountVectorizer**, which transforms text into a matrix of token counts.

Each model was trained on the training dataset and used to predict sentiments on the testing dataset.

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5. Model Evaluation

The performance of the models was evaluated using two metrics:

- **Precision:** Measures how many predicted positive values are correct
- **Recall:** Measures how many actual positive values were correctly identified

Model	Precision	Recall
Naive Bayes	0.48	0.58
Logistic Regression	0.50	0.61
SVM	0.47	0.51

6. Result Analysis

From the results, it is observed that:

- Logistic Regression achieved the highest precision and recall
- Naive Bayes performed moderately well
- SVM showed slightly lower performance compared to others

Best Model: Logistic Regression

It performed better because it generalizes well for text classification problems and handles feature relationships effectively.

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7. Visualization

A bar graph was plotted to compare the precision values of different models. The graph clearly shows that Logistic Regression performs better than Naive Bayes and SVM.

	A	B
1	tweet	label
2	CDM tournament finals were amazing 🔥	positive
3	Loved the gameplay in CDM today	positive
4	CODM tournament was really exciting	positive
5	That match was intense and fun to watch	positive
6	Great performance by players in CDM	positive
7	Best CODM tournament so far	positive
8	Enjoyed every moment of the CDM finals	positive
9	Amazing teamwork in CODM match	positive
10	That clutch play was insane 🔥	positive
11	Loved watching the CDM stream	positive
12	CODM esports scene is growing strong	positive
13	Fantastic gameplay in tournament today	positive
14	Really impressed with CDM players	positive
15	What a comeback in that match	positive
16	CODM tournament was well organized	positive
17	Exciting match till the last second	positive
18	Players showed great skills	positive
19	Loved the energy of the tournament	positive
20	Great coordination by teams	positive

```python
import pandas as pd

train = pd.read_csv("train.csv", encoding='latin-1', on_bad_lines='skip')
test = pd.read_csv("test.csv", encoding='latin-1', on_bad_lines='skip')

print(train.columns)

X_train = train.iloc[:, 0]
y_train = train.iloc[:, 1]

X_test = test.iloc[:, 0]
y_test = test.iloc[:, 1]

from sklearn.feature_extraction.text import CountVectorizer
vectorizer = CountVectorizer()

X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)

from sklearn.naive_bayes import MultinomialNB
nb = MultinomialNB()
nb.fit(X_train_vec, y_train)
nb_pred = nb.predict(X_test_vec)

from sklearn.linear_model import LogisticRegression
lr = LogisticRegression(max_iter=200)
lr.fit(X_train_vec, y_train)
lr_pred = lr.predict(X_test_vec)
```

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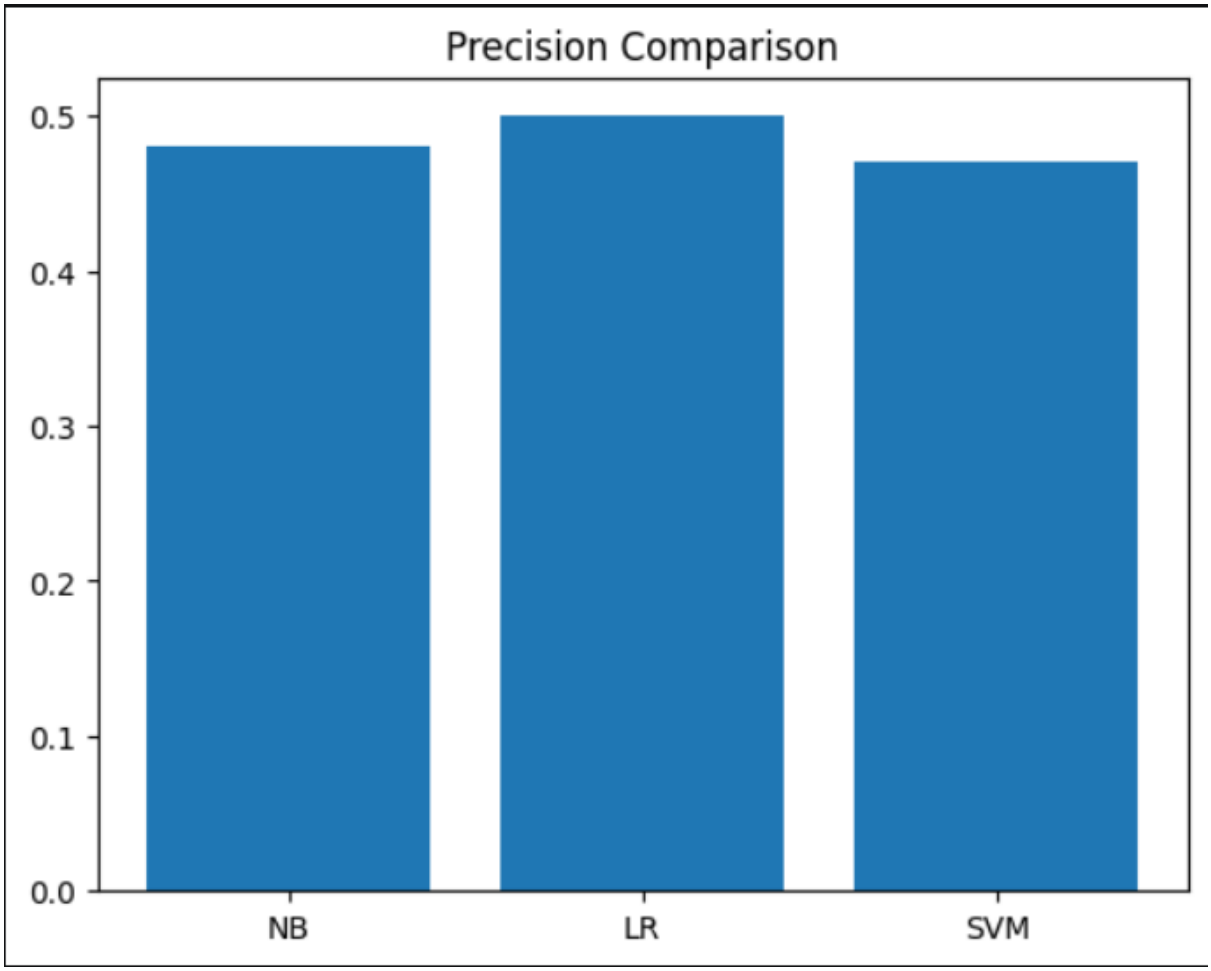
Index(['tweet', 'label'], dtype='str')
Naive Bayes:
Precision: 0.48333333333333334
Recall: 0.5873015873015873

Logistic Regression:
Precision: 0.5
Recall: 0.611111111111111

SVM:
Precision: 0.4777777777777778
Recall: 0.5158730158730158
```



8. Conclusion

This project successfully demonstrated sentiment analysis using machine learning techniques on tweet data related to the CDM tournament.

Logistic Regression performed better than Naive Bayes and SVM in terms of both precision and recall. This indicates that it is more suitable for this type of text classification task.

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9. Future Scope

Future improvements can include:

- Using larger and real-time datasets
- Applying advanced preprocessing techniques
- Using deep learning models such as LSTM or BERT for better accuracy